Fluid-structure interactions with numerics for a smoothed augmented Lagrangian variational principle applied to a wave-energy device

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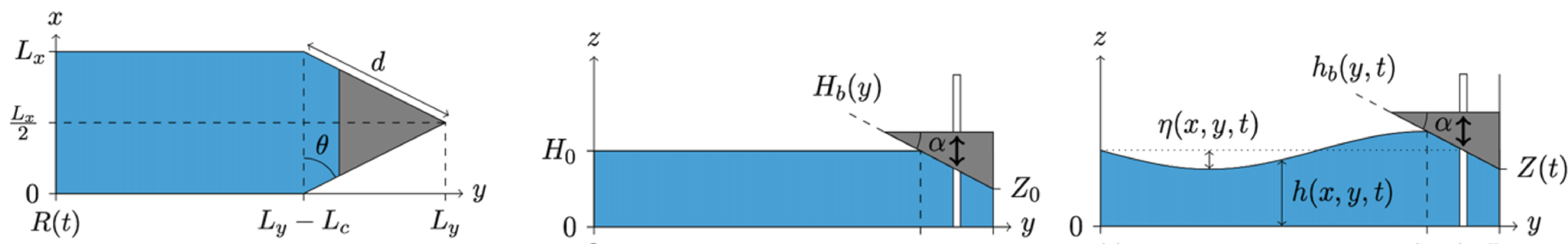
Introduction –Objective and FSI/Discretisation Strategy

Objective: We formulated, implemented & tested intriguing novel fluid-structure interaction (FSI) strategy for new wave-energy device with wave-buoy & buoy-generator interactions based on inequality constraints & Averaged Vector Field energy conservation. Used tedious bouncing ball model as pedagogically toy test. In five clear steps, our ***novel discretisation strategy*** is:

- S1: Write down the augmented Lagrangian with the inequality constraint embedded and derive the dynamics. *Why?* Includes all dynamics in one VP, directly with cunning coupling.
- S2: Approximate the Lagrange multiplier λ using new (implicit) function approximations to $\tilde{F}_+(\cdot)$, to solve for $\lambda(G)\le 0$ in terms of constraint $G\ge 0$. *Why?* Dynamics is stiff & geometric integrators perform poorly with (well-)known explicit functions $\lambda(G)$.
- S3: Write down new Lagrangian/Hamiltonian dynamics with λ eliminated. *Why?* Simpler & smoothed.
- S4: Use Average Vector Field energy (AVF) conservation as time integrator. *Why?* Working best to date. Symplectic integration failed.
- S5: Test! Derive/Obtain nonlinear solver stability criterion for Δt. *Why?* Get a handle on time-step size & convergence.



(a) Top view of the tank and buoy, outlining the tank's dimensions and how the buoy fits the shape of the contraction.

(b) Side view at rest ($t=0$), with the buoy constrained to move vertically.

(c) Side view at time t, with the buoy constrained to move vertically.

Figure 1. Sketches wave-energy device, buoy $h_B(Z(t),y)$. Courtesy Jonathon Bolton (dissertation in preparation).

S1,2,3: Models with inequality constraints

S1: Consider ***bouncing ball*** moving in vertical under constant gravity above $Z(t)\ge 0$, time t. It bounces with instant momentum reversal off the table. Augmented variational principle (VP) [2]:

$$0 = \delta\mathcal{L}_B[Z,W,\lambda] \equiv \delta\int_0^T W\dot{Z} - \frac{1}{2}W - Z - \frac{1}{2\gamma}\left(F_+(-\gamma Z-\lambda)^2-\lambda^2\right)\mathrm{d}t \tag{1}$$

with $\dot{Z}\equiv \mathrm{d}Z/\mathrm{d}t$, particle velocity $W(t)$, Lagrange multiplier $\lambda(t)$, constant $\gamma\gg 0$, and (maximum) function $F_+(Q)\equiv\max(0,Q)$ with $-Z\le 0$. Karush-Kuhn-Tucker (KKT) inequalities result from (2b): $-Z\le 0,\quad \lambda\le 0,\quad \lambda G(Z)=\lambda Z=0$. Variations of (1) subject to $\delta Z(0)=\delta Z(T)=0$ yield:

$$\delta W: \dot{Z} = W = \frac{\partial H}{\partial W},\qquad \delta Z:\ \dot{W} = -\frac{\partial H}{\partial Z} = -1+\tilde{F}_+(-\gamma Z-\lambda) = -1-\lambda \tag{2a}$$

$$\delta\lambda:\quad \lambda = -\tilde{F}_+(-\gamma Z-\lambda)\quad\text{with}\quad \tilde{F}_+(Q) = F_+(Q)F_+'(Q) = \max(0,Q). \tag{2b}$$

S2: Seek implicit and explicit approximations of $\lambda, \tilde{F}(Q), \tilde{F}(Q)$ to enforce stability ($0<b,d\ll 1$):

$$\text{Smooth}: \lambda_1(Z)\approx -ae^{-\gamma Z/b} = -\tilde{F}_+(Q=-\gamma Z-\lambda), Q = b\ln\left(\tilde{F}_+(Q)/a\right)+\tilde{F}_+(Q) \tag{3a}$$

$$\text{Cut}-\text{off}: \lambda_2(Z)\approx\min(0,\frac{Z}{b}), \tilde{F}_+(Q) = \frac{\max(0,Q)}{1+b\gamma},\quad F_+(Q) = \frac{1}{\sqrt{1+b\gamma}}\max(0,Q) \tag{3b}$$

$$\text{Asymp.}: \lambda_3(Z)\approx a\min\left(0,\frac{Z}{b}+\frac{Z|Z|}{d}\right), \tilde{F}_+(Q) = \max\left(0, Q+\frac{(d\Gamma-d\sqrt{\Gamma^2+4aQ/d})}{2a/\gamma}\right) \tag{3c}$$

$$C_2: \lambda_4(Z)\approx a\left\{\frac{Z}{b}, P(Z;\delta), 0\right\}\quad\text{with}\quad P(Z)\in C_2\text{–spline}, \tilde{F}(Q), F(Q)\text{ implicit}, \tag{3d}$$

with regions $\{Z\le-\delta, |Z|<\delta, Z>\delta\}$. S3: The integral of these forces $\lambda_i(Z)$ $(i=1,2,3,4)$ with respect to constraint variable $G=Z$ yields a potential $V_\lambda(G)$ in the augmented Lagrangian (1).

Wave-energy model

S1,S2,S3: Variational Boussinesq Model (VBM) water waves with surface velocity potential $\phi(x,y,t)=\phi_s(x,y,t)$ & depth $h(x,y,t)$ coupled to heaving buoy and power generator:

$$0 = \rho_0\delta\int_0^T\int_0^{L_x}\int_{R(t)}^{l_y(x)}\phi\,\partial_t h\,\mathrm{d}y\,\mathrm{d}x + MW\dot{Z} + (L_iI-K(Z))\,\dot{Q} - \mathcal{H}\,\mathrm{d}t \tag{4}$$

$$\begin{aligned}\equiv\rho_0\delta\int_0^T\int_0^{L_x}\int_{R(t)}^{l_y(x)}&\phi\partial_t h - \left(\frac{1}{2}h|\nabla\phi+h\psi\nabla h+\frac{1}{3}h^2\nabla\psi|^2+\frac{1}{6}h^3\psi^2+\frac{\beta}{90}h^5|\nabla\psi|^2\right)\\ &-gh(\frac{1}{2}h-H_0)-\frac{1}{2\gamma}\left(F_+(\gamma(h-h_b)-\lambda)^2-\lambda^2\right)\mathrm{d}y\,\mathrm{d}x-\int_0^{L_x}\dot{R}\left(h\phi+\frac{1}{3}h^3\psi\right)|_{y=R}\,\mathrm{d}x\\ &+MW\dot{Z}-\frac{1}{2}MW^2-MgZ+(L_iI-K(Z))\,\dot{Q}-\frac{1}{2}L_iI^2\mathrm{d}t,\end{aligned} \tag{5}$$

via constraint $G=(h_b-h)\ge 0$, with $\mathcal{H}$ total energy, $\beta=1$ for VBM and $\beta=0$ for Green-Naghdi model [4]. Variation of this VBM-VP with respect to variables $\mathbf{q}\equiv\{h,\phi,\psi,Z,W,Q,I\}$ yields the system of equations for (conservative part of) the entire wave-energy device ([1]):

$$\delta\phi:\quad \partial_t h = \frac{1}{\rho_0}\frac{\delta\mathcal{H}}{\delta\phi},\ \delta\psi:\ 0=-\frac{1}{\rho_0}\frac{\delta\mathcal{H}}{\delta\psi},\ \delta\lambda:\ 0=\frac{1}{\rho_0}\frac{\delta\mathcal{H}}{\delta\lambda},\ \delta h:\ \partial_t\phi = -\frac{1}{\rho_0}\frac{\delta\mathcal{H}}{\delta h} = -B+\lambda\approx -B+\underline{\underline{\lambda_i}} \tag{6a}$$

$$\delta W:\quad \dot{Z} = \frac{1}{M}\frac{\partial\mathcal{H}}{\partial W} = W \tag{6b}$$

$$\begin{aligned}\delta Z:\quad \dot{W}+\frac{\gamma_m}{M}G_I(Z)\dot{Q} &= -\frac{1}{M}\frac{\partial\mathcal{H}}{\partial Z} = -g+\frac{\rho_0}{M}\int_0^{L_x}\int_{R(t)}^{l_y(x)}F_+\left(\gamma(h-h_b)\right)F_+'\left(\gamma(h-h_b)\right)\mathrm{d}y\,\mathrm{d}x\\ &= -g-\frac{\rho_0}{M}\int_0^{L_x}\int_{R(t)}^{l_y(x)}\lambda\,\mathrm{d}y\,\mathrm{d}x\approx -g-\frac{\rho_0}{M}\int_0^{L_x}\int_{R(t)}^{l_y(x)}\underline{\underline{\lambda_i}}\,\mathrm{d}y\,\mathrm{d}x\end{aligned} \tag{6c}$$

$$\delta I:\quad \dot{Q} = \frac{1}{L_i}\frac{\partial\mathcal{H}}{\partial I} = I, \delta Q:\quad \dot{I}-\frac{\gamma_m}{L_i}G_I(Z)\dot{Z} = -\frac{1}{L_i}\frac{\partial\mathcal{H}}{\partial Q}-\underline{\mathit{dissipation}}\equiv\underline{-\frac{(R_i+R_c)}{L_i}I-\frac{I}{L_i|I|}V_s(|I|)}, \tag{6d}$$

with Bernoulli term B, added **electrical resistance/harvesting load** & λ_i–*approximation* (function of $h_B(Z(t),y)-h(x,y,t)$, cf. (3)). Nonlinear $K'(Z)\equiv G_I(Z)$ solves axi-symmetric 3D Maxwell's equations [1].

S5 testing & results: bouncing ball

Using $\lambda_3(Z)$ (3c) gives better convergence, since $F_+(Q)\to Q$ as $Q\to\infty$. Error is $O(\Delta t)^p$ with $p\approx 1$ compared with exact solution at $t\approx 3.5T$ for $a=10, b=d=0.1\Delta t^2, 10\Delta t^3$ (set-1, set-2).

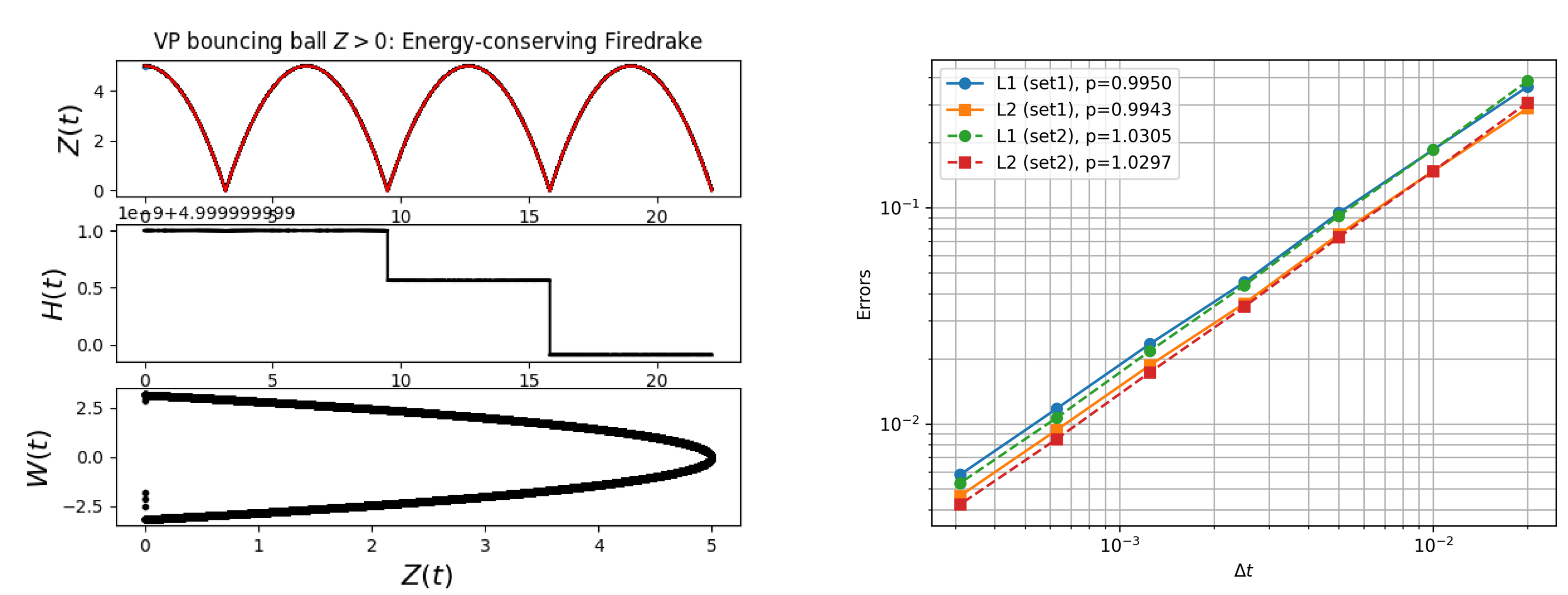


Figure 2. Left: Shown is ball under gravity bouncing off floor at $Z=0$. Δt is not tuned. Position vs. time (exact and numerical solutions shown), energy $H(t)$ fluctuation $\sim 10^{-9}$ around $H_0=5$ and phase diagram $\{Z^n,W^n\}$ of numerics. Right: errors; finest errors are L_1-norm: $(0.00582, 0.00532)$ & L_2-norm: $(0.00464, 0.00424)$ at $\Delta t=0.00031$.

Singularities in AVF have to be dealt with exactly for stability & accuracy, i.e. in Newton iterations:

$$\frac{(W^{n+1}-W^n)}{\Delta t} = -1-a\left(\frac{(Z_-^{n+1}+Z_-^n)}{2b}-\frac{({Z_-^{n+1}}^2+Z_-^{n+1}Z_-^n+{Z_-^n}^2)}{3d}\right)\begin{cases}1 & Z^{n+1}=Z^n\\ \frac{Z_-^{n+1}-Z_-^n}{Z^{n+1}-Z^n} & \text{else}\end{cases}.$$

S4 AVF energy-conservation strategy

Spectrally accurate Continuous Galerkin (CG) finite element methods were used in space (e.g., 2^{nd} order CG1, CG2, CG3) with time-efficient & error-reducing automatation via & implemented in *Firedrake* environment [5]. Key take-away is that energy conserving 2^{nd}-order AVF discretisation had to be used for stability, over time $t\in[t^n, t^{n+1}=t^n+\Delta t]$. Hence, we use linear-in-s variables $\mathbf{q}(s)\equiv\big(Z(s)=(Z^n+s(Z^{n+1}-Z^n), W(s)\big)$ & likewise for $\mathbf{q}(s)\equiv\{h(x,y,s),\phi(x,y,s),\psi(x,y,s),Z(s),W(s),I(s)\}$ (e.g., [3]):

$$\frac{\partial\mathbf{q}}{\partial t} = \frac{1}{\Delta t}\frac{\mathrm{d}\mathbf{q}}{\mathrm{d}s} = J(\mathbf{q}^{n+1/2})\int_0^1\frac{\partial H[\mathbf{q}(s)]}{\partial \mathrm{q}(s)}\mathrm{d}s, \tag{7}$$

with crucial exact integration. For polynomials this is automated in Firedrake and for the λ_i's this is derived/programmed due to difficult switches & limit $\mathbf{q}^{n+1}=\mathbf{q}^n$ in Newton iterations. J is skew-symmetric tensor for (2) (J constant) or (6) (J depends on $G_I(Z)$ in W, I-equations).

S5 Testing & results: wave-energy device

For our device, $\lambda_1(G)$ works poorly since exponential explodes and, hence, $\lambda_4(G)$ is best (using $a=g, b=10\Delta t^{3/2}$). Crucial C_2 smoothness in $\lambda_{1,4}$ is needed in CG-space. λ_4 is still too noisy.

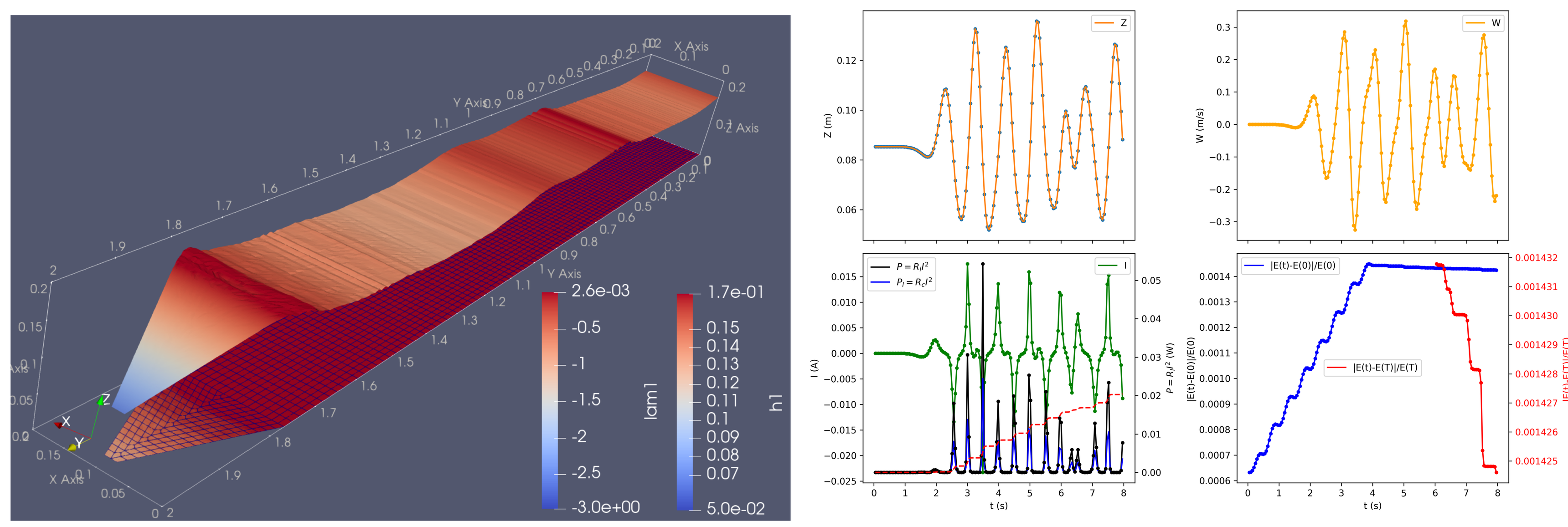


Figure 3. Left: snapshot free surface $h(x,y,t)$ & $\lambda_4(x,y,t)$ at $t=4.08$s on mesh. Right: $Z(t), W(t), I(t)$ & harvest/ circuit dissipation & scaled $\mathcal{H}(t)$ vs t. CG2 run. Post-processed λ_i still too noisy, but buoy motion & generator dynamics stable & smooth. Stable CG1–CG3 simulations completed. NB: $\delta\approx\Delta x$; buoy not shown, only its imprint.

Conclusions

The key take-aways of our new FSI-AVF framework for our new wave-energy device are:

- that singularities arising in AVF-integration, especially around the water- or contact line, need to be dealt with exactly; instant or continuous contact (C_2 needed) cases differ; and,
- that the free surface region needs to be solved exactly, i.e. have $\lambda=0$; both work in progress.

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